



High F_0 and musicianship make a difference: Pitch-shift responses across the vocal range



Brantly A. Sturgeon^a, Ryan J. Hubbard^b, Sara A. Schmidt^c, Torrey M. Loucks^{c,d,e,*}

^a Department of Electrical and Computer Engineering, University of Illinois, 619S. Wright St., Champaign, IL 61820, USA

^b Department of Psychology, University of Illinois, 603 E. Daniel St., Champaign, IL 61820, USA

^c Program in Neuroscience, University of Illinois, 505S. Goodwin Ave., Urbana, IL 61801, USA

^d Department of Speech and Hearing Science, University of Illinois, 901S. Sixth St., Champaign, IL 61820, USA

^e Beckman Institute for Advanced Science and Technology, 405N. Mathews Ave., Urbana, IL 61801, USA

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ABSTRACT

The control of vocal pitch plays a central role in speech and singing, where accurate tonal production relies on modulating F_0 production. Previous research has shown that vocal pitch is modulated in response to artificial pitch perturbations during ongoing vocalization (i.e., the pitch-shift response) with response characteristics differing between musicians and non-musicians. However, little is known about the behavior of these responses in the higher portion of a speaker's vocal range for amateur musicians or non-musicians. The present study addresses whether pitch control changes as target pitch is systematically increased and whether this variation is affected by musicianship. We predicted that as F_0 moves away from comfortable speaking level, pitch-shift response peak amplitude would increase and peak latency would decrease, while participants with musical experience would have lower amplitude responses than those without experience. Results showed that peak amplitude linearly increased as F_0 was raised, and that peak latency similarly decreased but with more variability. However, musical participants had larger pitch-shift response amplitudes and longer peak latency times than non-musical participants, which is an unanticipated finding for persons with musical training. These patterns suggest that pitch control can be manipulated by vocal tasks that vary F_0 while still being subject to strong modulation by musical background.

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1. Introduction

The control of voice fundamental frequency (F_0) has a central role in speech and singing in which F_0 variation conveys information on lexical content, speaker identity, emotional state, and intent (Juslin & Laukka, 2001; Lieberman & Michaels, 1962; Lima, Castro, & Scott, 2013). Of increasing interest to researchers is the control of voice F_0 in the higher portion of a speaker's range (Liu, Auger, & Larson, 2009; Liu & Larson, 2007; McCrea & Morris, 2005; Trainor, Marie, Bruce, & Bidelman, 2014). English speakers tend to remain in their low-to-mid F_0 range for everyday conversation, while volitional production of high voice F_0 is reserved for singing and other emotive and theatrical contexts; thus, a fundamental question is whether the control mechanisms for voice production differ at voice F_0 levels across an individual's speaking range. The pitch shift feedback paradigm has been used to explore control mechanisms during vocalization and has shown important evidence of differing responses at a high F_0 compared to a conversational F_0 (Liu & Larson, 2007). In this paper, we attempt to address this question more directly by testing pitch shift responses as F_0 is systematically varied. We have extended this question further by including participants with and without musical training to test the effect of musicianship on audio-vocal control across a wide F_0 range.

Physiologically, there is strong evidence that producing high F_0 vocalizations differs from producing low F_0 vocalizations (Gay, Hirose, Strome, & Sawashima, 1972; Hirano & Ohala, 1969; Hirano, Vennard, & Ohala, 1970; Liu, Behroozmand, Bove, & Larson, 2011). In general, activity of the laryngeal muscles increases as F_0 during vocalization increases; this is particularly true of the cricothyroid and vocalis muscles, whose activity increases dramatically at higher frequencies (Chhetri, Neubauer, & Berry, 2012;

* Corresponding author at: Department of Speech and Hearing Science, University of Illinois, 901 S. Sixth St., Champaign, IL 61820, USA.

E-mail addresses: brantly106@gmail.com (B.A. Sturgeon), rjhubba2@illinois.edu (R.J. Hubbard), saschmi3@illinois.edu (S.A. Schmidt), tloucks@illinois.edu (T.M. Loucks).

Kochis-Jennings, Finnegan, Hoffman, & Jaiswal, 2012; Titze, 1994), although the extrinsic laryngeal muscle, sternohyoid, also displays a contribution to higher frequencies as the larynx is raised (Hirano & Ohala, 1969). Higher F_0 production is also associated with increases in tracheal pressure (Verdolini-Marston, Titze, & Druker, 1990), as well as increases in loudness and glottal airflow (Isshiki, 1965). Vocal training seems to produce physiological differences as well; along with general changes in laryngeal activity, vocal training may be associated with different patterns of vocal fold control, leading to greater control in vocal F_0 (Gay et al., 1972; Herbst, Qiu, Schutte, & Švec, 2011). This physiological evidence indicates that voice control operates differently at higher F_0 levels and likely varies with singing ability.

A useful paradigm for investigating vocal control mechanisms that is now widely recognized is the pitch shift response paradigm. In seminal studies of this paradigm (Bauer & Larson, 2003; Burnett, Senner, & Larson, 1997; Burnett, Freedland, Larson, & Hain, 1998; Hain, Burnett, Larson, & Kiran, 2001; Liu et al., 2009; Liu & Larson, 2007), participants receive pitch-shifted auditory feedback during vocalizations, and vocal responses are recorded. Compensatory responses are typically observed, where the participant's pitch decreases in response to an upward shift and increases in response to a downward shift (Burnett et al., 1997). Following responses, in which the vocalized pitch changes in the same direction as the stimulus, are observed less frequently (Behroozmand, Korzyukov, Sattler, & Larson, 2012); these inverted response types suggest the possible involvement of two neural mechanisms in vocal production, though research on this is currently ongoing. One model of vocal control posits that an internal mechanism regulates the output vocal frequency by comparing an “efferent copy” of the vocalized F_0 against the desired frequency, which is usually conversational F_0 (Hain et al., 2001). However, at frequencies outside of the comfortable conversational range, the mechanism may not have a typical target F_0 to influence the magnitude and direction of responses, which could result in different patterns of responses. This, as well as the physiological differences at higher frequencies, warrants investigation of pitch-shift responses at different F_0 levels.

Two previous studies have investigated pitch-shift responses at higher vocal frequencies (Liu & Larson, 2007; Liu et al., 2009). Results showed greater pitch shift response magnitudes and shorter response latencies at higher F_0 levels. These results suggest that control mechanisms may differ during high F_0 vocalizations; namely, higher amplitude and faster responses could reflect more efficient processing or active monitoring. However, these interpretations are hindered somewhat by design limitations; specifically, the stimulus levels were chosen somewhat arbitrarily (a conversational level and a “much higher level”), and only one frequency level was tested for each category. To date, there has been no research on pitch-shift responses across the vocal range on an individual subject basis.

Pitch shift responses that vary with F_0 could also be influenced by vocal training. In general, when musically trained participants are told to ignore pitch-shifted feedback and continue vocalizing at the same pitch, they show greater suppression of the pitch-shift response than musically naïve participants (Zarate & Zatorre, 2005, 2008; Zarate, Wood, & Zatorre, 2010). Singers also show faster compensatory vocal reactions with quicker returns to baseline following a pitch-shift than non-singers (Behroozmand, Ibrahim, Korzyukov, Robin, & Larson, 2014), have a lower threshold for pitch-shift responses (smaller shift stimuli), and tend to show greater after-effects when the altered auditory feedback is removed, suggesting specific tuning of their internal model for F_0 control (Jones & Keough, 2008; Keough & Jones, 2009). However, it is still unclear whether pitch-shift responses of singers differ at higher vocal frequencies. Moreover, general musicianship or musical ability has yet to be tested. Individuals with general musical training are more familiar with varying the pitch of their instruments and, thus, across the vocal range we predict these trained individuals could show distinctive audio–vocal control compared to non-musicians.

The current study attempts to address the important knowledge gaps in pitch range and musicianship by examining how pitch-shift responses differ across a broad range of systematically varied F_0 targets that include higher and lower vocal frequencies. We hypothesized that response magnitude would increase and latency would decrease as voice F_0 increased as per previous studies (Liu & Larson, 2007), with the additional prediction that pitch-shift responses would be smaller below conversational F_0 due to reduced tension in vocal fold musculature. Following previous research on singers, we predicted that the pitch-shift response magnitude would be lower for musically trained individuals across the vocal range—especially at higher F_0 levels—as musically trained individuals are better able to suppress pitch shifts in order to maintain accurate pitch matching. These findings have relevance for understanding how pitch control is maintained when F_0 is altered that could hold relevance both for language production and vocal control in singing.

2. Materials and methods

2.1. Participants

Twenty-one female young adults who are students at the University of Illinois participated in the experiment, either for course credit or for payment (mean = 22 years, range 18–30 years). All participants passed a hearing screening at 25 dB hearing level bilaterally at 250, 500, 1000, 2000, and 4000 Hz and none reported perceived hearing loss. Participants were asked two questions regarding musical training: whether they had taken band or choir while in school (from kindergarten until 12th grade, K-12) and if they had private music lessons outside of school. All participants that had taken vocal lessons outside of K-12 school were placed in the musical group. Participants with private training on other instruments were placed in the musical group if this training lasted more than 2 years. Additionally, all participants performed an online tone discrimination task (Mandell, 2009) and all were able to differentiate two tones apart at least 12.5 Hz. The mean discrimination for the musical group of 4.7 (sd 2) Hz was lower than the non-musical

Table 1
Individual subject information listed by group.

Subject	Age (years)	TD (Hz)	Musical experience		F ₀
			K-12	Extra-curricular	
Non-musical group					
NM1	30	1.2	Percussion 2 years	Piano 1 year	213
NM2	24	7.2	Band 2 years	Piano <1 year	212
NM3	21	19	Band 1 years	None	203
NM4	18	8.3	None	None	189
NM5	24	16.8	Choir 2 years	None	230
NM6	20	2.1	Choir	Piano 2 years	220
NM7	21	0.9	Band 1 year	None	182
NM8	19	7.8	None	Piano 2 years	208
NM9	21	3.9	Band 4 years	None	177
NM10	19	9.6	Choir 2 years	Piano 1 year	170
Musical group					
M1	26	7.2	Band 10 years	Piano 17 years	215
M2	18	3.5	Band 9 years	9 years piano & voice	223
M3	20	6.3	Vocal 8 years	Multiple instruments >5 years	242
M4	20	3.9	Band >5 years	11 years piano & voice	226
M5	26	9	Band 3 years	Voice multiple years	180
M6	26	3.3	Choir 5 years	Piano & vocal multiple years	207
M7	24	2.7	Band 8 years	9 years piano & voice	215
M8	28	1.95	Band >5 years	9 years violin & flute	218
M9	20	3	Band 4 years	8 years piano	179
M10	21	6	Band 6 years	9 years piano & clarinet	211
M11	20	5	Band/Choir 4 years	10 years clarinet, piano & choir	216

TD – tone discrimination; K-12: musical experience in school; Extra-curricular: musical training outside school; F₀ – average fundamental frequency during sentence repetition.

group mean of 7.5 (sd 6.5) Hz, but the difference was not statistically significant. All participants used English as their dominant language and none were native/fluent speakers of a tonal language. Detailed subject information including age, musical training information and scores on the tone discrimination task is included in Table 1. All participants provided informed consent and the experiment was approved by the University of Illinois Institutional Review Board.

2.2. Apparatus

During the entire experiment, participants were seated in a sound booth and wore Sennheiser HD-280 Pro headphones and a head-worn Shure microphone placed 2 cm away from the corner of the mouth. They were asked to vocalize the vowel /a/ at a conversational loudness level. Feedback regarding loudness level was presented to each speaker using a Dorough Loudness Monitor (model 40-A) placed in front of them. The participants were told to maintain their loudness within the yellow area of the loudness monitor to ensure that their average volume was around 70 dB. The vocal signal from the microphone was amplified with a Mackie mixer (model 1202) and shifted in pitch with a digital signal processor (Eventide Ultra Harmonizer H7600). MIDI software (MAX/MSP v.5 by Cycling 74, Walnut, CA) connected to a MOTU (model UltraLite-mk3 Hybrid) controlled the timing, duration and magnitude of the pitch shifts via the Eventide ultra-harmonizer. The general headphone output setting was approximately 80 dB SPL, adjusted slightly to achieve a loud but comfortable setting for each participant. Pink masking noise, which equalizes energy across octaves, giving a more even perception of equal loudness was additionally presented to the subject at 60 dB during each trial to mask bone conduction. The voice signal was lowpass filtered using an 8-pole Butterworth filter (Frequency Devices 9002, filter model LP09) with a 2.49 kHz cutoff to avoid aliasing and recorded using data acquisition software (DATAQ Instruments WinDaq Pro ver. 3.36, Akron, OH) along with TTL timing signals.

2.3. Procedures

Following the initial setup, all participants read five Harvard sentences (phonetically balanced sentences; Rothauser et al., 1969) twice each in a comfortable speaking tone while vocal pitch was measured using the pitch tracking module within the KayPENTAX CSL Model 4500. The average F₀ of these sentences was taken to be the participant's comfortable speaking F₀ (hereafter conversational F₀) and subsequently used to estimate the target pitch stimuli for each participant. The conversational F₀ value was entered into a custom MATLAB script (MathWorks ver. 2011b, Natick, MA), which generated tone stimuli based on the Rosenberg glottal pulse model (Rosenberg, 1970) for the simulated vowel /a/ at 9 target pitch levels relative to the participant's conversational F₀: 2 semitones below conversational F₀, at conversational F₀, and 6 semitones above conversational F₀. Importantly, this range was used to capture F₀ levels that varied above and below the individual's conversational F₀ with the focus on frequencies above F₀ levels typical for conversation (Stemple, Stanley, & Lee, 1995). Participants were instructed and trained to imitate the vowel /a/ to accurately match the frequency of the target tone stimuli on each trial. All participants indicated that they were able to comfortably imitate the vowel at all of their target pitches. Each participant's conversational F₀ is listed in Table 1.

Prior to the experiment, participants practiced five /a/ imitative vocalizations to become familiar with the pitch-matching experimental paradigm. On each trial of the actual experiment, the participant heard the target frequency for three seconds through the headphones. The participant then attempted to match the target pitch by continuously vocalizing an imitation of the vowel /a/ for five seconds. An LED light placed in front of the participant indicated when to start and stop vocalizing. During the five second vocalization period, four pitch shift stimuli were played at random intervals. The onset of the first pitch-shift stimulus occurred between 500 ms and 700 ms after the beginning of vocalization, and the interstimulus interval varied between 900 and 1000 ms. Each pitch shift stimulus lasted 100 ms and had a magnitude of 100 cents. These experimental manipulations were determined to be effective for eliciting pitch-shift responses by a previous methodological study (Bauer & Larson, 2003). To reduce the possibility of stimulus expectancy by the participants, the direction of the pitch-shift stimulus was randomized and control trials containing no shifts were randomly distributed throughout the experiment; thus, the experiment contained equal numbers of upward shifts, downward shifts, and control trials in a randomized order for each F_0 target (semitone level) to eliminate expectancy. During a run for an F_0 target, each participant produced a total of 24 vocalizations containing 4 stimuli; as such, 96 responses were recorded for each F_0 target while providing short breaks between each run.

2.4. Data analysis

All data were analyzed using customized routines for event-related averaging techniques in MATLAB. First, the voice signal from the continuous recordings was parsed into individual trials of one second in length. Each trial captured the following for a single pitch shift stimulation: a 200 ms baseline prior to the onset of the stimulus, the 100 ms stimulus period, and a post-stimulus response period of 700 ms. Using the Sawtooth Waveform Inspired Pitch Estimator Prime (SWIPE') algorithm (Camacho, 2007), fundamental frequency estimates were formed using a 5 ms sample period with a pitch strength threshold of 0.06 to reject poor pitch estimates due to insufficient intensity. These estimates were then filtered by a 5-point median filter to reject spurious estimates. Fundamental frequency waveforms in Hz were converted to measurements in cents by using the 200 ms baseline period as a reference point. Downward pitch shift responses were reversed ($y = -x$) so upward and downward responses could be compared directly. Artifacts and trials with interruptions or non-vocal utterances were rejected through a two-step process. First, differences in cents between two adjacent time points in a waveform were compared to identify sharp discontinuities. Differences in cents between the mean of the waveform and the most extreme values were calculated and any values of 75 cents or greater than the mean resulted in exclusion of the trial. Finally, these estimates were filtered a second time using a 5-point binomial filter to soften quantization effects while maintaining high frequency information. On average, 10% of the trials were rejected from each participant.

The remaining trials remaining were averaged together using event-related processing techniques. The averaging window included the 200 ms pre-stimulus period, the 100 ms stimulus duration and the 700 ms post-stimulus period. Average waveforms were calculated separately for upward shifts, downward shifts, and control conditions for each subject. Additionally, based on previous research indicating that following responses may differ from compensatory responses (Behroozmand et al., 2012), these two types of responses were analyzed separately. Before extracting dependent variables, the five averaged waveforms from each participant were smoothed with a 40-point Gaussian filter.

2.5. Dependent variables

The pitch-shift responses were primarily evaluated by determining the peak amplitude and peak latency of compensatory and following responses. The positive value of peak amplitude for all responses was always used so responses could be compared regardless of shift direction. The first peak, which corresponds to the timing of the first response for this mechanism, was analyzed exclusively for both responses types. It is based on a temporal window that captures the first peak latency of pitch shift responses (approximately 200–700 ms). The proportion of compensatory and following responses was compared to determine whether the task elicited differences in response types. To evaluate how well participants performed the pitch-matching task, F_0 accuracy was compared by calculating the root-mean-square error of actual F_0 vs. F_0 of the target pitch during control tasks.

Each variable was analyzed separately using analysis of variance. The factors included for the comparisons of peak amplitude and peak time were two within-subject factors of target pitch (9 levels) and pitch shift direction (2 levels: Up/Down) and a between-subject factor of group (2 levels: Musical/Non-musical). For the analyses with pitch shift direction as a factor, compensatory responses were assessed separately from following responses and vice versa. An alpha level of $p = 0.05$ was used to evaluate statistical significance for all comparisons in this study. Post-hoc tests with error correction were applied if a significant main effect of target pitch or an interaction with target pitch was detected.

For the analyses comparing compensatory with following responses, upward F_0 responses were compared separately from downward F_0 responses. The particular ANOVA models included the within subject factors of target pitch (9 levels) and response type (2 levels: Compensatory/Following) and a between-subject factor of group (2 levels: Musical/Non-musical). Then the proportion of compensatory responses vs. following response types was compared after taking the average of the upward and downward shifts for compensatory and following responses separately. This ANOVA model included the within subject factors of target pitch (9 levels) and response type (2 levels: Compensatory/Following) and the between-subject factor of group (2 levels: Musical/Non-musical). F_0 accuracy was also compared across target pitch (9 levels) and between the two groups (Musical/Non-musical) with ANOVA. Appropriate post-hoc tests (Sidak test, corrected to $p < 0.05$) were conducted when a significant effect or interaction with target pitch

was detected. Lastly, linear regression models were determined to assess the dependence of pitch-shift response amplitude on target pitch. A similar approach was used to assess the dependence of peak latency on target pitch.

3. Results

Compensatory and following responses to the pitch-shift stimuli were evident in each participant. In Fig. 1, compensatory pitch-shift responses are shown in a waterfall format for a single participant from the non-musical and musical groups (downward responses plotted as upward by taking $y = -x$ for downgoing responses). For both, variability in compensatory peak amplitude is evident, but a general increase with target pitch is evident for each direction with downward responses generally having greater amplitude than upward responses. Similar waterfall representations are shown in Figs. 2 and 3 for the groups as a whole showing response peaks in the 200–300 ms range. The non-musical group (Fig. 2) shows compensatory responses that increase as target pitch increases. Downward compensatory responses were associated with greater peak amplitude than upward compensatory responses. The following responses in both directions have early latency peaks but amplitudes are smaller than the compensatory responses. Compensatory responses in the musical group are readily identifiable (Fig. 3) and have peak amplitudes that appear greater for the highest target pitches. The early latency following responses are similarly smaller than compensatory responses but the first peaks of the following responses appear distinctly.

3.1. F_0 accuracy

As per Fig. 4a, tracking RMS error of the control trials changed significantly over the 9 target pitch levels ($F(1,19)=5.1$, $p<0.05$, $\eta^2=0.21$). RMS error was significantly lower for the conversational F_0 condition relative to the highest target pitch (Sidak, $p<0.05$). Importantly, the main effect comparing accuracy between the groups was not significant so further comparisons were not conducted ($F(1,19)=0.81$, $p>0.05$, $\eta^2=0.01$).

3.2. Proportion of responses

The distribution of the proportion of compensatory pitch-shift responses and the proportion of following responses is shown across target pitch level in Fig. 4b. This distribution indicates that the probability of observing a compensatory response tends to increase with target pitch while the probability of following responses tends to decrease with target pitch. Across the range of targets, the proportion of compensatory responses was significantly greater than following responses (response type: $F(1,19)=91.9$, $p<0.001$,

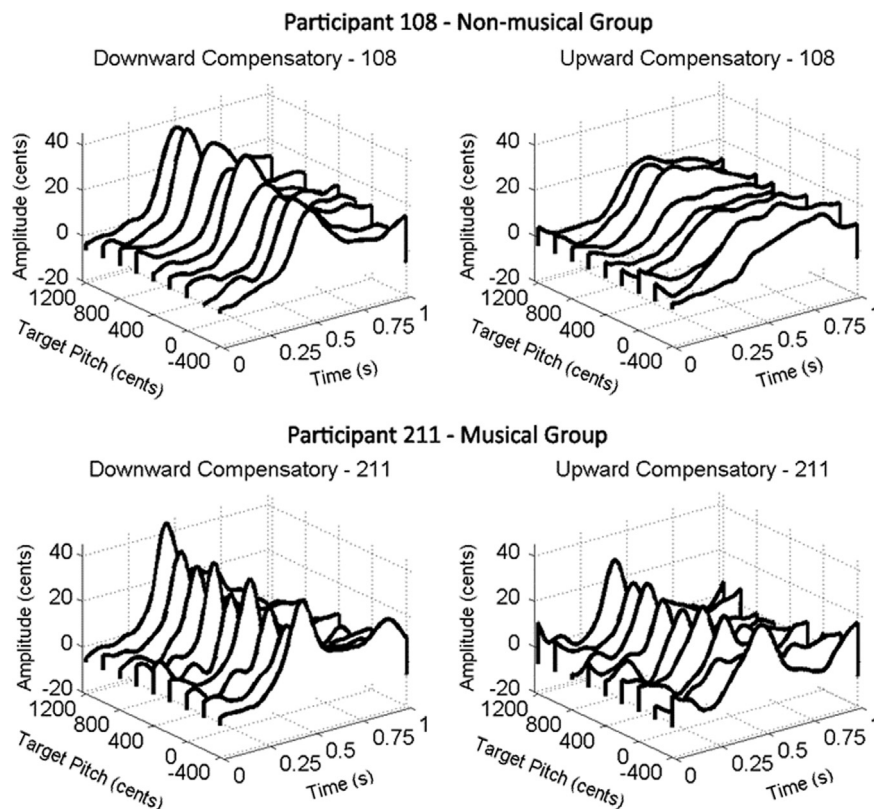


Fig. 1. Waterfall plots showing average compensatory pitch shift feedback responses in two representative subjects. Each plot shows the average response at each Target pitch level. The X axis shows time (s), the Y axis shows target pitch (0 represents conversational F_0), and the Z axis shows response amplitude (cents).

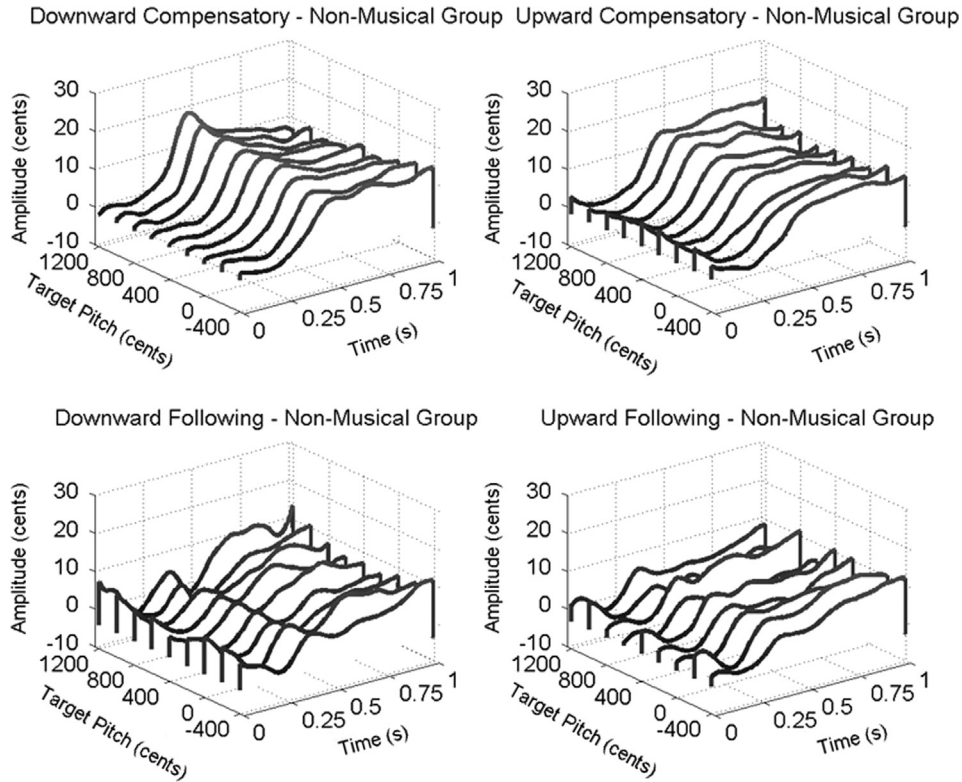


Fig. 2. Waterfall plots showing group averages of the compensatory and following responses in non-musical participants: X axis – time (s); Y axis – target pitch (cents); and, Z axis – amplitude (cents). The upper two plots show compensatory responses to upward and downward shift directions while the lower plots show the corresponding following responses.

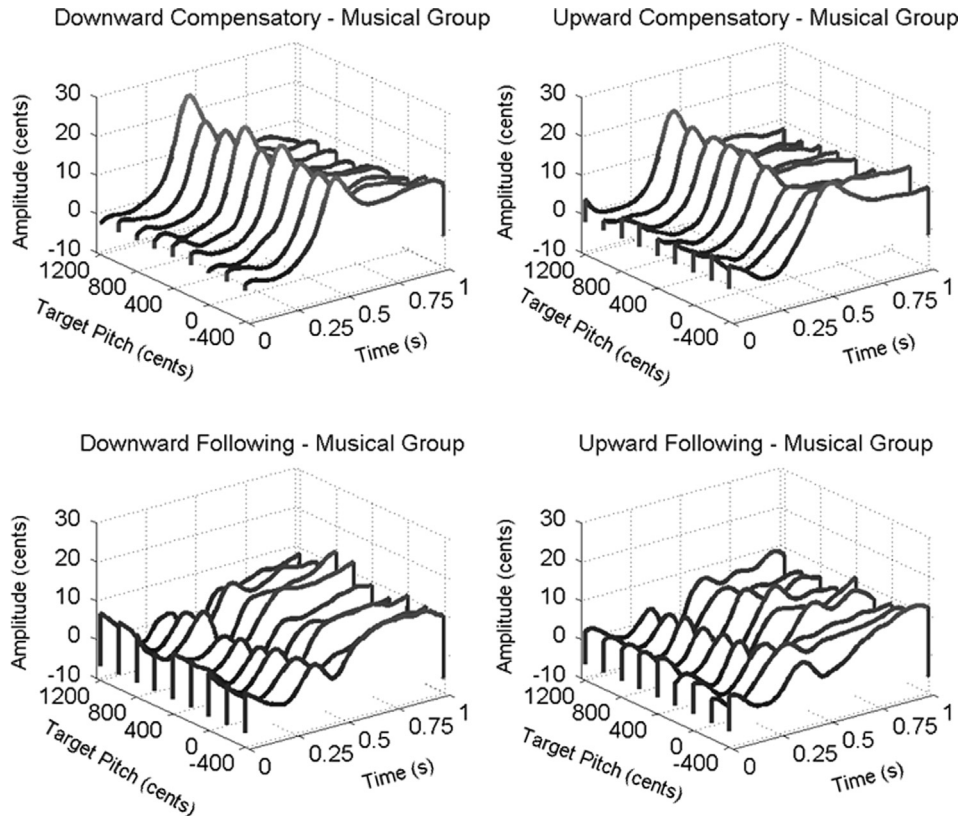


Fig. 3. Waterfall plots showing group averages of the compensatory and following responses in musical participants: X axis – time (s); Y axis – target pitch (cents); and, Z axis – amplitude (cents). The upper two plots show compensatory responses to upward and downward shift directions while the lower plots show the corresponding following responses.

$\eta^2=0.83$). The main effects of musicianship ($F(1,19)=1.7$, $p>0.05$, $\eta^2=0.02$) and target pitch were not significant on the proportion of responses ($F(1,19)=1.3.7$, $p>0.05$, $\eta^2=0.01$). However, the interaction between response type and target pitch were significant in terms of a linear trend ($F(1,19)=18.7$, $p<0.001$, $\eta^2=0.49$) and a quadratic trend ($F(1,19)=14.7$, $p<0.01$, $\eta^2=0.42$) where the target

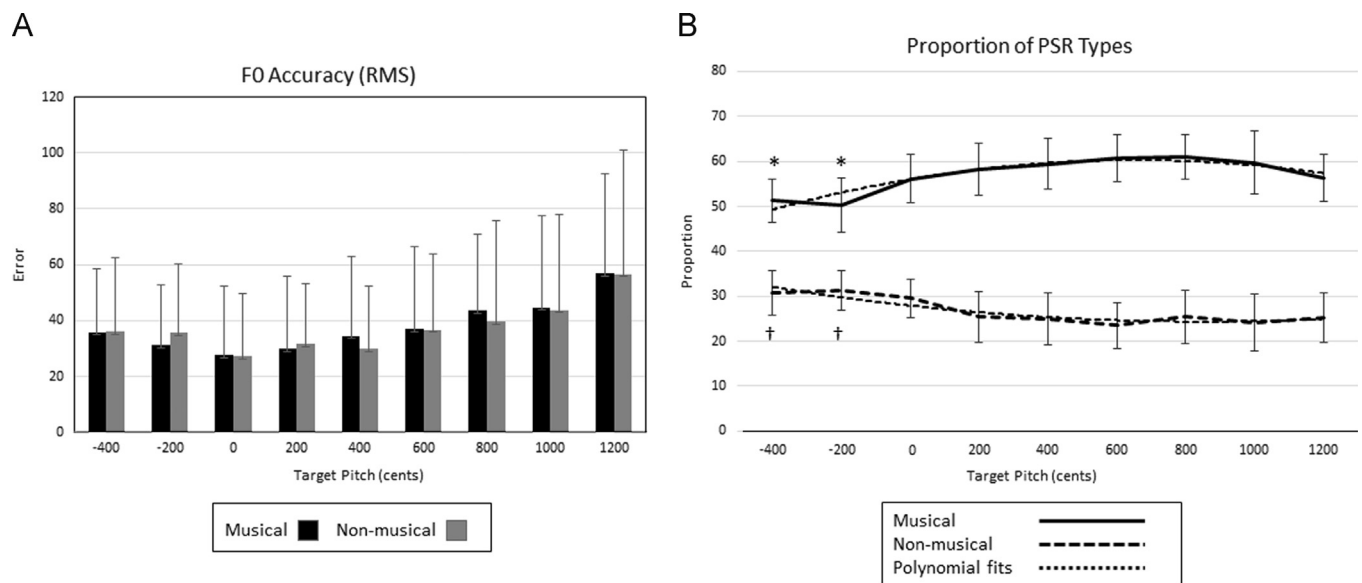


Fig. 4. (A) Depicts average group accuracy in matching the target tone, measured as root-mean square error (RMS) at each Target pitch. (B) Depicts the proportion of compensatory vs. following responses across both groups at each target pitch. The light dotted lines show polynomial fits to each curve. * indicate a significant post-hoc difference ($p < 0.05$ corrected, Sidak) between the particular Target pitch and the highest Target pitch level for compensatory responses. The cross represents a corresponding significant post-hoc effect for following responses ($p < 0.05$ corrected, Sidak). The vertical bars represent one standard deviation in both figures.

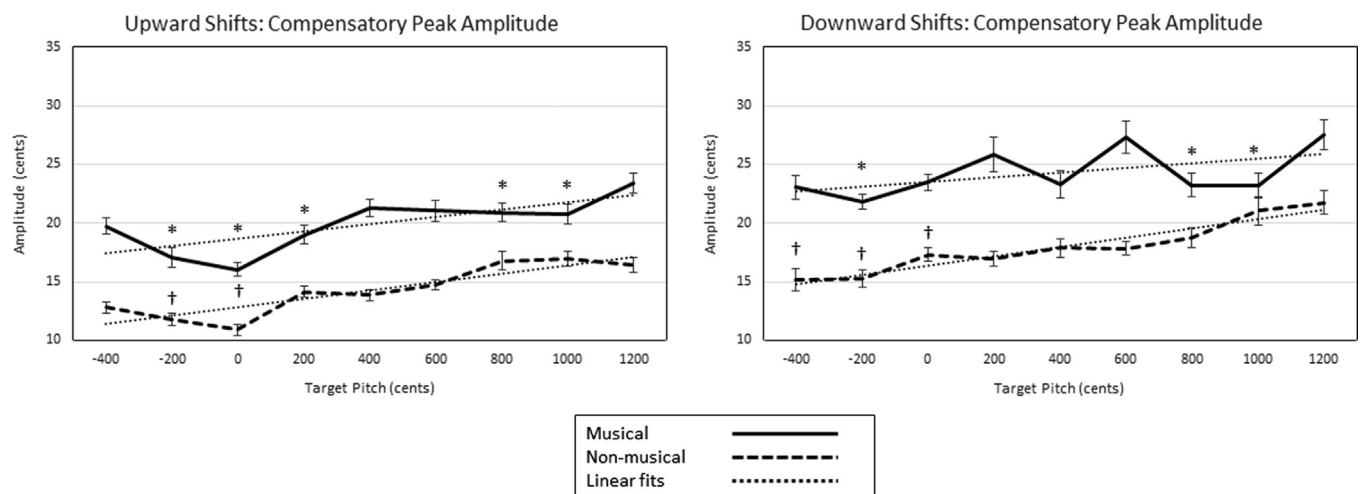


Fig. 5. Plots depict compensatory response amplitudes (cents) in musicians and non-musicians across the target pitch levels. The left plot shows upward shift responses, and the right plot shows downward shift responses. The light dotted lines are linear fits for each of the conditions. In both figures, * indicates a significant post-hoc difference ($p < 0.05$ corrected, Sidak) between the particular target pitch level and the highest target pitch level for the musical group. The cross represents a corresponding significant post-hoc effect for the non-musical group ($p < 0.05$ corrected, Sidak). The vertical bars represent one standard deviation in both figures.

pitch is used to predict response proportions. Sidak post-hoc tests indicated the proportion of responses was significantly greater at the highest target pitch than the two lowest target pitches ($p < 0.05$ corrected, see Fig. 4b).

3.3. Pitch-shift response peak amplitude

The peak amplitude results of the compensatory pitch-shift response are presented before following responses. Downward pitch shifts elicited significantly larger compensatory responses than upward pitch shifts as shown by the main effect of direction ($F(1,19) = 8.9$, $p < 0.01$, $\eta^2 = 0.81$). Compensatory pitch-shift response amplitude also increased significantly across pitch level as indicated by a significant linear effect of target pitch ($F(1,19) = 9.9$, $p < 0.01$, $\eta^2 = 0.85$). The musical group showed significantly higher amplitude compensatory responses across upward and downward shifts (group: $F(1,19) = 5.7$, $p < 0.05$, $\eta^2 = 0.62$), but an interaction between group and the other factors was not detected.

The change in compensatory pitch-shift response F_0 across target pitch for musically experienced and inexperienced groups is shown in Fig. 5. Given significant group differences with a linear pitch effect, post-hoc corrected comparisons were evaluated separately for each group. The pattern of post-hoc differences with respect to the highest target pitch as displayed in Fig. 5 indicates that pitch-shift response amplitudes at lower target pitches tend to be significantly reduced relative to the highest target, but not

uniformly. The linear trends that were evaluated for each pitch-shift direction and group indicate a significant positive slope for amplitude as target pitch increases for certain conditions as displayed in Fig. 5. The following lists the regression coefficients and correlations for each pitch-shift response amplitude–target pitch relationship where A = pitch-shift response amplitude and TP = target pitch: (a) Musical – Upward pitch-shift response: $A = 0.74TP + 19.2$, ($r(11) = 0.87$, $p < 0.05$); (b) Non-musical – Upward pitch-shift response: $A = 0.63TP + 11.3$, ($r(10) = 0.94$, $p < 0.05$); (c) Musical – Downward pitch-shift response: $A = 0.63TP + 23.6$, ($r(11) = 0.72$, $p > 0.05$); and (d) Non-musical – Downward pitch-shift response: $A = 0.56TP + 15.8$, ($r(10) = 0.98$, $p < 0.05$).

The contrast between compensatory and following responses is shown in Fig. 6A with pitch-shift response types averaged across upward and downward shift directions. The statistical comparisons between compensatory and following response amplitudes were conducted separately for each pitch shift response type. For upward responses, compensatory amplitude was significantly greater than following amplitude (response type: $F(1,19) = 52.6$, $p < 0.001$, $\eta^2 = 0.73$). A significant interaction between response type and group indicated that group differences were present for compensatory responses but not following responses (response type: group: $F(1,19) = 8.4$, $p < 0.01$, $\eta^2 = 0.31$). A response type by target pitch interaction indicated that compensatory amplitude varied with pitch while following amplitude did not (response type: target pitch: $F(1,19) = 8.4$, $p < 0.01$, $\eta^2 = 0.79$). For the downward responses, the only difference detected was response type (compensatory vs. following: $F(1,19) = 42.2$, $p < 0.001$, $\eta^2 = 0.69$) indicating downward compensatory responses had greater amplitude than downward following responses. The statistical comparison between upward following and downward following response amplitude did not detect main effects or interactions.

3.4. Pitch-shift response peak latency

Similar to peak amplitude, the peak latency of the compensatory pitch-shift response changed as target pitch increased, but tended to decrease as target pitch increased. Fig. 7 illustrates the trends for each group and direction. Overall, a significant linear

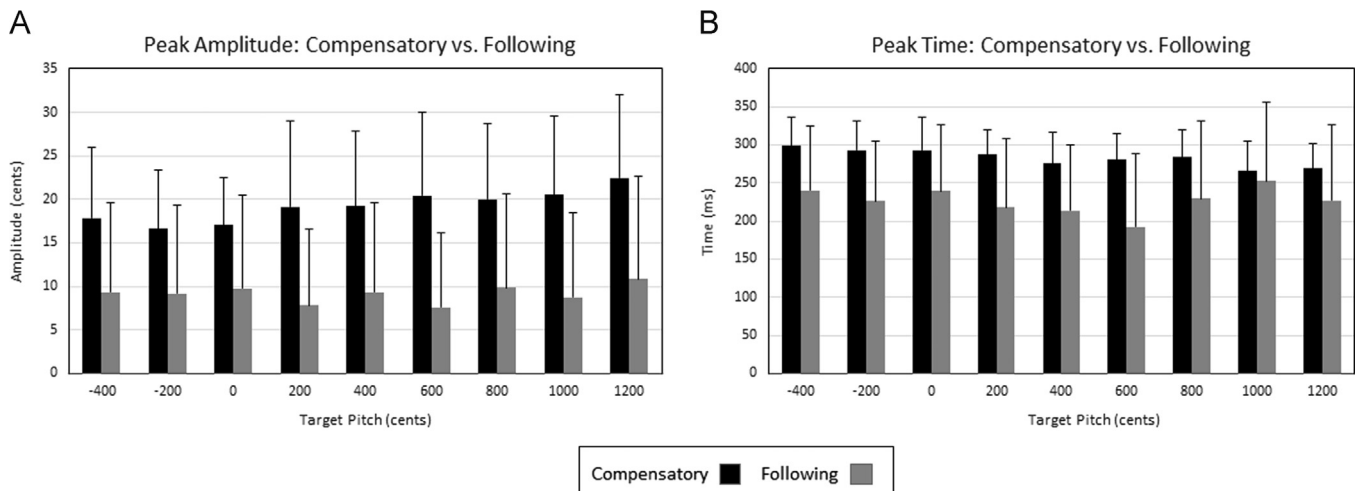


Fig. 6. Bar plots comparing average peak amplitude and average latency between compensatory and following responses across the target pitch levels (averaged across groups). In both figures, the vertical bars represent one standard deviation. (A) Depicts peak amplitudes and (B) depicts peak latencies.

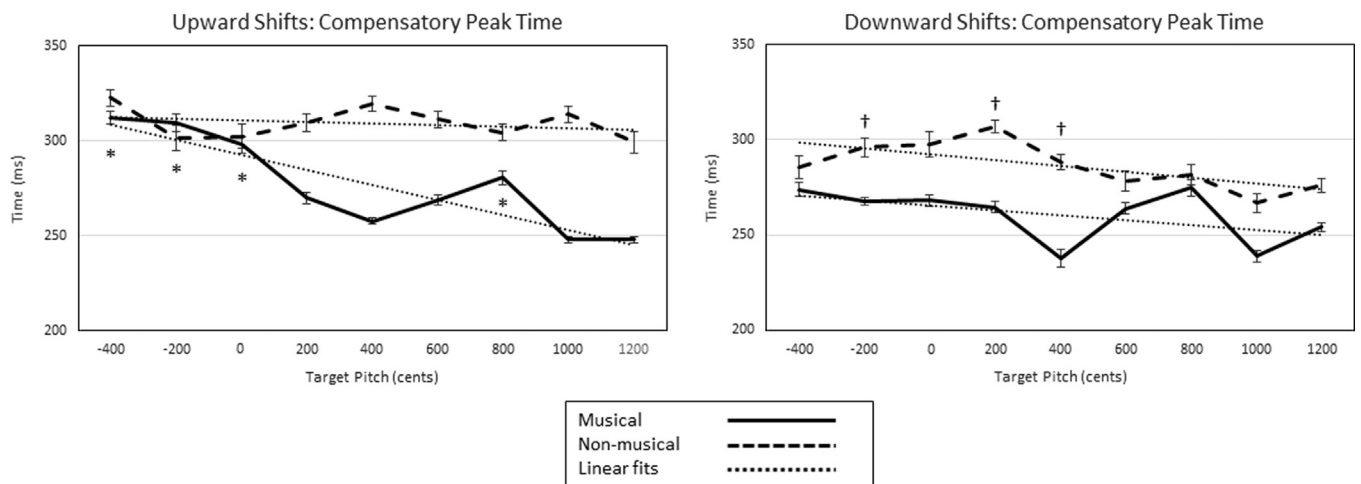


Fig. 7. Plots depicting compensatory response latencies (ms) in musicians and non-musicians across Target pitch levels. In both figures, the vertical bars represent one standard error. The left plot shows upward shift responses, and the right plot shows downward shift responses. The light dotted lines show linear fits for each of the conditions. * indicates a significant post-hoc difference ($p < 0.05$ corrected, Sidak) between the particular target pitch level and the highest target pitch level for the musical group. The cross represents a corresponding significant post-hoc difference for the non-musical group ($p < 0.05$ corrected, Sidak).

effect for target pitch ($F(1,19)=9.3$, $p<0.01$, $\eta^2=0.82$) indicates peak time generally decreased as target pitch increased. The effect of shift direction was significant with upward shifts having earlier peak latencies than downward shifts ($F(1,19)=11.6$, $p<0.01$, $\eta^2=0.89$) across both groups. Also similar to the amplitude results, the musical group had lower peak latencies than the non-musical group ($F(1,19)=7.6$, $p<0.05$, $\eta^2=0.74$).

The linear trends that were evaluated for each compensatory pitch-shift direction and group are presented in Fig. 7. The following lists the regression coefficients and correlations for each pitch-shift response peak time–target pitch relationship where T =pitch-shift response peak time and TP =target pitch: (a) Musical – Upward pitch-shift response: $T=-7.0TP+316.2$, ($r(11)=-0.94$, $p<0.05$); (b) Non-musical – Upward pitch-shift response: $T=-0.8TP+313.8$, ($r(10)=-0.53$, $p>0.05$); (c) Musical – Downward pitch-shift response: $T=-2.6TP+273.8$, ($r(11)=-0.71$, $p>0.05$); and (d) Non-musical – Downward pitch-shift response: $T=-3.1TP+274.2$, ($r(10)=-0.82$, $p<0.05$).

The contrast between upward compensatory and following peak latencies indicated that upward following response peaks occurred significantly earlier than compensatory response peaks ($F(1,19)=18.6$, $p<0.01$, $\eta^2=0.52$). The parallel comparison for downward shifts indicated that downward following peak latencies were significantly earlier than compensatory latencies ($F(1,19)=8.0$, $p<0.05$, $\eta^2=0.32$). Fig. 6b shows this contrast between compensatory and following responses, with pitch-shift response responses averaged across upward and downward shift directions. The statistical comparison between upward following and downward following latencies did not detect any differences.

4. Discussion

This is the first pitch-shift study to employ a broad and systematic set of F_0 targets in a pitch-shift feedback paradigm. The range of pitch targets covered more than one octave, including targets lower than a typical speaking F_0 and a high F_0 target one octave above typical F_0 . The initial expectation (Hypothesis 1) was that the extreme high and low F_0 targets would produce different pitch-shift response characteristics from the conversational F_0 target, in that larger amplitude responses would be observed, but instead there was a strong linear dependence between increasing F_0 and the compensatory pitch-shift response, while the amplitude and latency of responses below conversational F_0 were not different from patterns found for conversational F_0 . While the amplitude of the compensatory pitch-shift response increased with target pitch for the musically trained and untrained groups, the musical group did not show relative response suppression as we initially predicted (Hypothesis 2). Instead, the musical group showed higher response magnitude than the non-musical group and, although not predicted, the musical group also had shorter peak latencies, which point to a more sensitive laryngeal control system in amateur musicians.

4.1. Pitch-shift response characteristics vary with target pitch

In this study, our initial expectation was that the extreme high and low F_0 targets would differentiate pitch-shift response characteristics, but instead there was a strong dependence linked to increasing F_0 for the compensatory pitch-shift response while the amplitude and latency of responses below conversational F_0 were not different from patterns found for conversational F_0 . A linear trend was found indicating compensatory pitch-shift response amplitude increased with equal interval increases in cents, which was evident in both groups. Evidence of a similar trend for decreasing pitch-shift response peak latency with equal interval increases in cents was also present, though not as robust. A linear relationship is, of course, an over-simplification because the cents scale is a log 2 scale. We do not intimate that the pitch-shift/ F_0 response is strictly linear; particularly, given the abundant evidence from self-triggered responses, suppression and laryngeal anesthesia studies (Liu, Behroozmand, & Larson, 2010; Larson, Altman, Liu, & Hain, 2008; Zarate & Zatorre, 2008). Rather, the current data indicates that compensatory pitch-shift response amplitude and peak latency are partly a function of a speaker's target F_0 . In clear contrast, the amplitude and latency of following pitch-shift responses did not vary with target pitch, which builds on previous evidence that these responses differ in nature (Behroozmand et al., 2012). This current findings replicate Liu and Larson (2007), as well as Liu et al. (2009), while extending the results across a significant portion of the vocal range.

The divergence between compensatory and following response characteristics with increasing F_0 was also observed in response proportions. Across the range of targets, significantly more compensatory responses were observed with the proportion of compensatory response increasing with target pitch above conversational F_0 while following response proportions decreased. The result of greater compensatory response proportions at high F_0 levels parallels the two previous studies of the pitch-shift mechanism at low and high frequencies (Liu et al., 2009; Liu & Larson, 2007). It remains unclear why increasing pitch would lead to a greater proportion of compensatory responses; it is possible that the increased involvement of the cricothyroid and vocalis muscles is responsible, though further investigation will be required to understand this relationship. Overall, When the F_0 target is explicitly controlled, compensatory responses are the expected response and accordingly determine meaningful pitch-shift response variation.

Our study differs from previous work in that we did not vary vocal loudness. The large number of target pitch conditions took approximately one hour, which was composed of frequent vocalizations across much of the duration. Thus, manipulating loudness in addition to F_0 would have prolonged the experiment and might have fatigued participants. All participants were instructed to vocalize at a conversational loudness level, which was consistently followed. Our results are therefore most relevant for typical loudness levels but clearly support a predictable relationship between target pitch level and PSR characteristics, which could not be specified

in previous studies. A potential limitation is that our study used imitative tasks rather than spontaneous vocalizations, so the tasks were neither explicit speech nor singing tasks. In this initial study, we considered it important that the participants were able to perform the tasks in a similar way so that the group comparisons would be meaningful. Considering that we found effects both for F_0 variation and musical background, the paradigm can be meaningfully extended to study more naturalistic vocalizations.

The current findings appear to support a meaningful relationship between the state of muscular tension in the vocal folds and modest changes in pitch-shift response characteristics. This follows from the [Liu et al. \(2011\)](#) study, in which clear contractions of the internal cricothyroid and thyroarytenoid muscles were observed during a compensatory pitch-shift response, particularly when participants were vocalizing in falsetto. The high degree of muscle tension in falsetto that lengthens and stiffens the vocal folds reveals that laryngeal muscles can actively contract to produce the pitch-shift response. Laryngeal muscular tension patterns in that study were more ambiguous below falsetto, in that contractions and relaxations were observed, but these results do not preclude systematic muscle adjustments. Thus, the state of the vocal folds is relevant in modulating the pitch-shift response; however, as has been shown in numerous studies, the pitch-shift response is clearly mediated by central nervous system mechanisms that are subject to varied influences, including musical experience.

4.2. Musical experience

Our study is the first to show, perhaps surprisingly, larger amplitude responses and shorter latencies of responses in musically trained participants compared to non-musicians. As our starting point however, we predicted the musical participants in the current study would show diminished pitch-shift responses compared to non-musical participants because past studies emphasize suppression of pitch-shift response responses by vocally trained musicians ([Zarate & Zatorre, 2008](#); [Zarate et al., 2010](#)), even though relative suppression could diminish with longer exposure to the stimuli ([Jones & Keough, 2008](#)).

If control mechanisms of the voice are altered through training or experience, then vocally trained musicians are the most likely to show pitch-shift response characteristics that reflect these influences and correspondingly differ from musically naïve speakers. The divergent pitch-shift response characteristics in singers, particularly the suppression, shown in several studies ([Behroozmand et al., 2014](#); [Zarate & Zatorre, 2008](#); [Zarate et al., 2010](#)) presumably arises from central neural mechanisms that reduce the gain of the pitch-shift response allowing the vocalist to maintain a target note more effectively than a non-musical person. But we were interested in whether suppression would also be evident in amateur musicians with varied training. If these individuals also showed suppressed pitch-shift response amplitude across the target pitch range, it would provide evidence that distinctive vocal control in musicians is not limited to vocal training.

Although the musicians were all amateurs, we also found distinct evidence for altered pitch-shift response characteristics in this group. However, the results were in the opposite direction from our prediction – namely, all musically trained individuals showed pitch-shift response amplitudes that were higher or equal to the mean of the non-musical group. The difference in amplitude was mirrored by shorter pitch-shift response peak latencies in the musicians as well. These differences were evident across the target pitch levels, but were most pronounced for high F_0 targets. The target that was most similar between the groups was our typical conversational F_0 , suggesting it is necessary to compare these groups at more than one target pitch.

In general, the range of F_0 targets in this study especially above conversational F_0 could have required increased vocal effort and monitoring of vocalization that lead to adjustments in the auditory–motor system towards a more corrective mode, i.e., a stronger feedback driven mode. This could entail higher pitch-shift response gain associated with a monitoring system that is generally envisioned as an internal model within the central nervous system. If a more active or engaged audio–vocal system was evidenced by the larger amplitudes and shorter latencies above conversational F_0 , it was more pronounced for the musical group. Increased sensitivity in amateur musicians leading to greater pitch-shift gain might result from more active monitoring of task performance. There is previous pitch-shift response evidence that response gain can be modified by instructions ([Hain et al., 2000](#)). By specifically instructing participants to voluntarily shift their pitch in response to the stimulus, listeners can modulate the amplitude of the response, though this amplitude increase was constrained to a secondary feedback response following the primary pitch-shift response. While the pitch-shift response paradigm does not confirm that the musically experienced participants were exhibiting more active control, this reasoning is consistent with musicians being able to engage experience dependent brain mechanisms for audio–vocal control and task-dependent modulation of the pitch-shift response ([Zarate et al., 2010](#)). This was previously shown to be localized in the left hemisphere where cortical modulation of the pitch-shift response was solely observed in musicians with absolute pitch ([Behroozmand et al., 2014](#)). Although we did not test for absolute/relative pitch or neural responses, we suggest that our findings are consistent with amateur musicians being able to exhibit distinct and flexible audio–vocal control.

A possible counter-explanation of group differences is that the musicians were performing the matching task more consistently (i.e. more accurately matching the target tone). If so, then inconsistent performance in the non-musical group might account for the pitch-shift response differences but would also complicate valid group comparisons. However, the RMS error measure indicated both groups had similar consistency in matching the target pitch across the range of F_0 targets. If both groups were on average showing similar consistency in performance, it is unlikely that simple variability in musically naïve participants' performance explains our group differences. Instead, it seems that musically trained participants were more sensitive to the perturbation apart from the baseline influence of increased vocal fold tension. Additionally, the current study did not use musical notes as targets which meant that F_0 targets did not have any greater familiarity for musicians as opposed to the non-musicians. Potentially, musically trained individuals would have shown lower amplitude responses if the targets were musical notes, while the non-musical would have shown the same pattern.

An alternative consideration of why the musical group showed increased gain rather than suppression is that studies have varied in the form of instructions (Zarate & Zatorre, 2005, 2008; Zarate et al., 2010). In certain studies, participants were instructed to either ignore the pitch-shift stimulus, or to purposefully compensate in the opposite direction to the shift. The musically trained participants showed nearly complete suppression of the pitch-shift response when told to ignore the stimulus, whereas musically naive participants still showed a response. In contrast, no group differences in response amplitude were found in the compensate condition. The recent study by Behroozmand et al. (2014) that compared musically trained and naive participants with a pitch-shift response paradigm found response timing differences but not magnitude differences, which additionally indicates that suppression is not a universal finding in musician/non-musician comparisons. In our experiment, participants were told to focus on accuracy without specific instructions regarding how to react to the stimulus. Thus, it may be the case that individuals with musical training are more proficient at actively suppressing pitch-shift responses, but also show larger pitch-shift responses when the task is to maintain a target pitch. It seems the combined evidence from previous studies and the current results now demonstrate that musical individuals maintain a broader dynamic range of audio–vocal control evinced by more flexible pitch-shift response characteristics. Further investigation is still necessary to fully understand the potent effects of musical training on vocal control, but pitch-shift response testing remains an important tool for testing audio–vocal interactions that can be employed across a range of F_0 levels.

4.3. Is high F_0 special?

As presented in the introduction, speaking or singing at higher frequencies than typical speaking F_0 involves higher effort and laryngeal muscle changes that occur with greater precision with training. High F_0 is therefore not a typical range for most speaking tasks and is usually considered to be the domain of singing. Our results introduce new information that audio–vocal control also shows distinct characteristics at high F_0 . Previous studies employing the pitch-shift feedback paradigm have not systematically explored how audio–vocal integration functions at higher F_0 levels. The results of the present study clearly show that pitch-shift response peak magnitude increases as vocal F_0 increases, while peak latency systematically decreases—evinced that faster, greater F_0 responses can be engaged for tonal control. These important findings were strongly conditioned by musical training. Taken together, these results suggest that audio–vocal control mechanisms are flexible and may operate differently above an individual's typical speaking range. Our results with the musical group additionally highlight that musical training modulates audio–vocal control mechanisms in a frequency dependent manner. Such distinctions in a basic audio–vocal mechanism may not be restricted to singing. Tonal languages that demand a higher F_0 range for certain lexical distinctions may also contribute to experience dependent effects on pitch-shift responses (see Liu, Xu, & Larson, 2010 for an example of how native Mandarin speakers show distinct pitch-shift response attenuation). We only demonstrated this frequency dependence for young adult females; however, our interesting findings should be further investigated across the full vocal range particularly in trained vocalists but additionally in males, other age groups and using varied loudness levels while not neglecting the F_0 range below typical speaking F_0 .

5. Summary

As target pitch was shifted upwards across 9 levels that exceeded an octave range, peak amplitude increased while peak latency (~ response time) decreased reflecting an increased sensitivity to the stimuli. This pattern was clear in both non-musicians and amateur musicians, but in contrast to our prediction, amateur musicians were distinguished by having even larger amplitude responses than non-musicians. Increased gain in musicians, rather than the typical suppression, supports an interpretation that musicians have a wider dynamic range of responses and a more flexible audio–vocal system. We expect this pattern to be operative in both singing and speech, but it requires further investigation particularly in professional singers and other languages.

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